EMBEDDED SYSTEMS CASE STUDY

STM32-Based Soil Irrigation and Environmental Monitoring System

Automatic soil-moisture monitoring with temperature, humidity and Wi-Fi status reporting

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1. Problem Statement

Plants require an appropriate amount of water for healthy growth. Manual watering can be irregular, time-consuming and may lead to overwatering or underwatering. This project presents a microcontroller-based soil irrigation and environmental monitoring system. It reads the soil moisture level, monitors ambient temperature and humidity, and provides status information through a Wi-Fi interface.

2. Objectives

- Build the monitoring and control prototype around an STM32F401CCU6 (Black Pill) microcontroller.
- Read the analog output of a soil-moisture sensor using the STM32 ADC.
- Use a DHT11 sensor to measure temperature and relative humidity.
- Indicate the soil condition locally; the prototype logic uses an LED indication and a buzzer alert where configured.
- Use an ESP8266 Wi-Fi module to transmit sensor/status information for viewing on a mobile device.
- Apply a threshold-based decision to identify dry soil and initiate the irrigation response.

3. Proposed Solution

The STM32F401CCU6 acts as the main controller. The soil-moisture sensor is connected to an analog input, and its output is sampled by the ADC. The firmware compares the reading with a calibrated threshold. When the soil is classified as dry, the system indicates the condition and can activate the irrigation output; when the soil is sufficiently moist, the irrigation output remains inactive. A DHT11 provides temperature and humidity readings. The ESP8266 provides Wi-Fi communication so that sensor values and system status can be viewed remotely.

The project notes specify PA0 for the soil-moisture analog signal, PB8 for the DHT11 data signal, PB9 for the buzzer, and PA2 as the STM32 receive input connected to the ESP8266 transmit output. The exact relay/pump driver wiring and any additional indicator pins should be confirmed against the final assembled circuit.

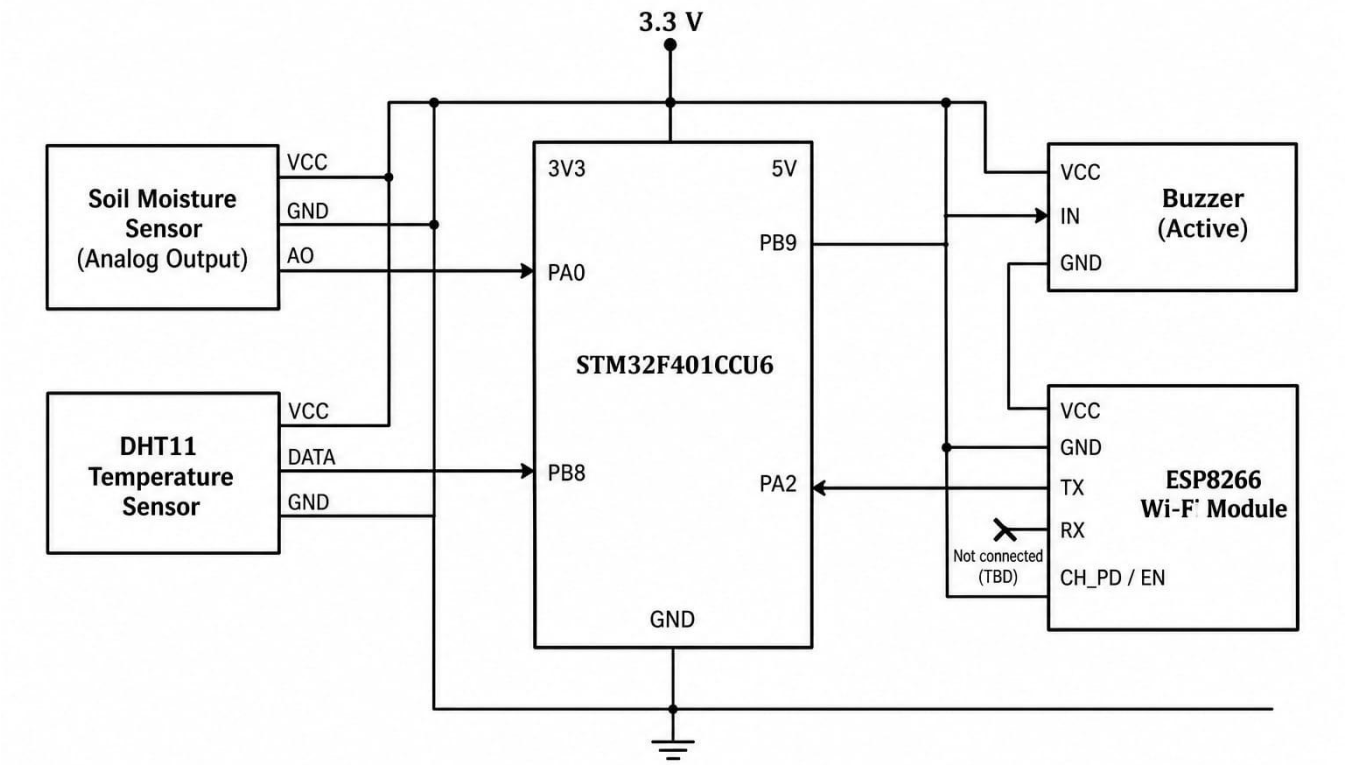
4. System Working

1. Power-on: initialise the STM32 GPIO, ADC, timing and serial communication peripherals.
2. Soil sensing: sample the analog voltage from the soil-moisture sensor through PA0.
3. Decision: compare the ADC reading with the calibrated dry-soil threshold. The threshold must be determined from actual dry and wet soil readings.
4. Local indication: indicate wet soil with the configured status LED; indicate dry soil using the configured blinking/alert behaviour and buzzer, if connected.
5. Environmental sensing: read temperature and humidity from the DHT11 connected to PB8.
6. Wi-Fi reporting: send the measured values and soil status through the ESP8266 so they can be monitored on a phone or browser.
7. Repeat: continue sampling and updating the status at a suitable interval.

5. Circuit Design

The prototype is assembled on a breadboard using jumper wires. The photograph supplied for this report shows the controller boards, a breadboard, connected sensor/interface modules and a buzzer-style module. The ESP8266 development board is visible near the lower part of the breadboard, while a second controller board is mounted near the upper-right area. The photograph documents the physical prototype; it does not by itself establish every pin connection or confirm the presence of a water pump/relay.

For a practical pump implementation, the pump must be driven through a correctly rated relay or transistor/MOSFET driver and a separate suitable power supply. The pump must never be powered directly from an STM32 GPIO pin. Use a common ground where required by the driver/interface design, and keep water away from exposed electronics.



6. Pin Mapping

Pin / Interface	Function	Connection / Purpose
PA0	ADC input	Analog output (AO) of soil-moisture sensor
PB8	Digital GPIO	DHT11 data signal (as specified in the project wiring notes)
PB9	GPIO output	Buzzer control
PA2	USART2_RX	Receives serial data from ESP8266 TX
ESP8266 Wi-Fi	UART / Wi-Fi	Sends monitoring information to a mobile/browser interface
Pump control	Driver output	Relay or MOSFET driver for the water pump; confirm final pin assignment
LED indicator	GPIO output	Local wet/dry status indication; confirm final pin assignment

Note: The pin mapping above follows the project connection details available for this prototype. The final firmware and wiring should be checked together before hardware demonstration.

7. Prototype

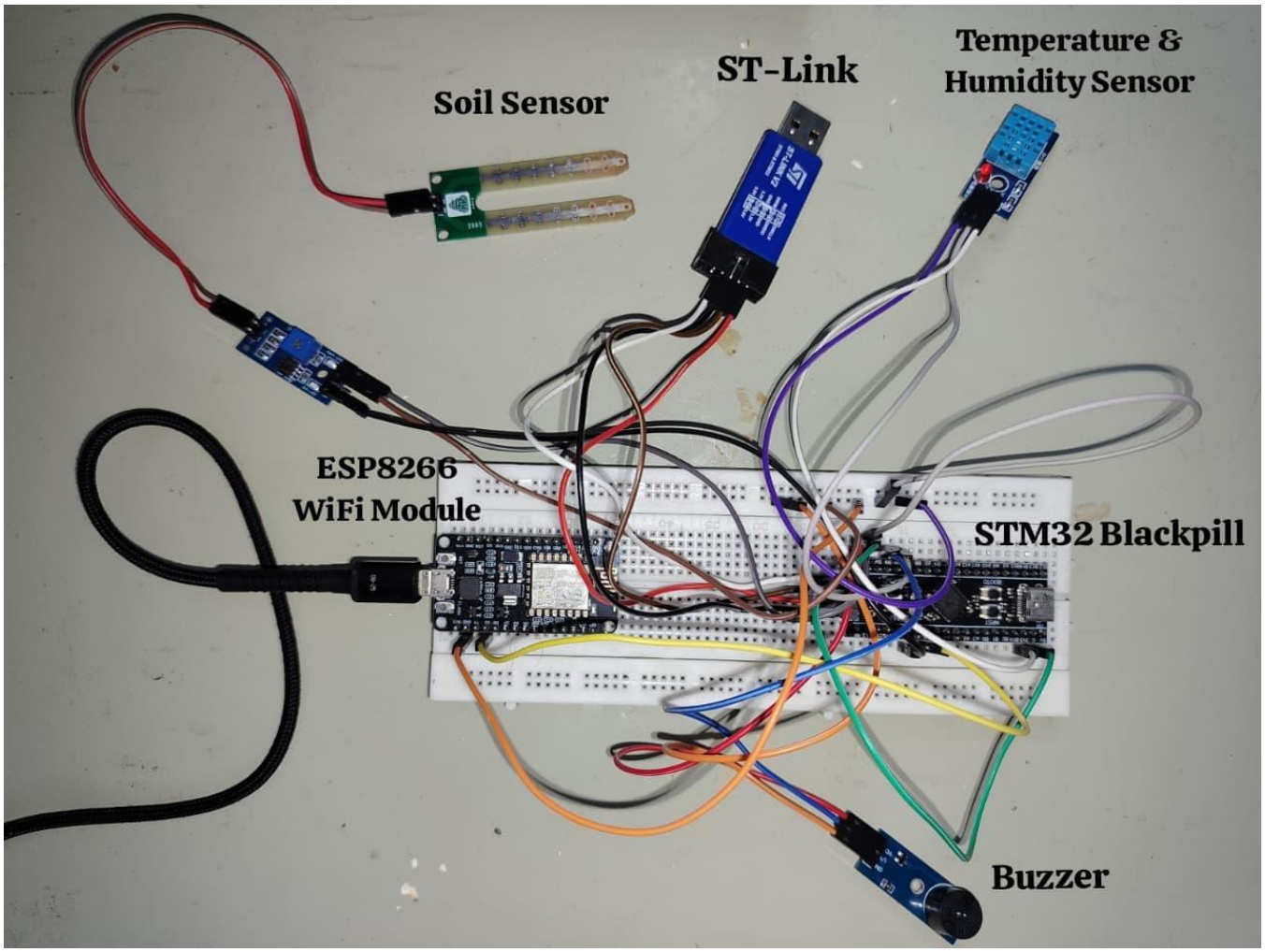


Figure 1. Breadboard prototype of the soil irrigation and environmental monitoring system.

The prototype photograph shows the connected development boards and modules arranged on a breadboard, with USB power/programming cables and jumper-wire connections. The image is included as the project hardware evidence supplied by the team.

8. Testing Plan

- Soil sensor calibration: record ADC readings in dry soil and moist soil, then select a threshold between the observed ranges.
- Dry-soil response: verify that the dry condition is detected and the configured irrigation/alert output is activated.
- Wet-soil response: verify that the wet condition is detected and the irrigation output is inactive.
- DHT11 check: compare displayed temperature and humidity readings with a reference thermometer/hygrometer if available.
- Wi-Fi communication: verify that the ESP8266 transmits updated sensor values and status to the monitoring interface.
- Repeated operation: run multiple wet/dry transitions and check for unstable switching; introduce hysteresis or averaging if required.

Test results should be entered after measurements are performed. No numerical performance results are claimed in this report because measured test data were not supplied.

9. Limitations and Future Work

Limitations: soil-moisture readings vary with soil type, sensor placement and sensor condition. A fixed threshold may not work equally well for every plant or soil. DHT11 readings have limited accuracy compared with higher-grade environmental sensors. Wi-Fi reporting depends on network availability. The supplied photograph does not show a clearly identifiable pump-and-driver assembly, so automatic water delivery should be verified separately.

Future work: calibrate the sensor for different soil types, add hysteresis to prevent rapid pump switching, integrate a water-level sensor, add a proper pump driver and enclosure, log readings over time, and build a mobile-friendly dashboard with alerts. A solar-powered supply and scheduled irrigation modes can also be explored.

10. Acknowledgement

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